

オウ イシン
王 一臻

WANG Yizhen

Master | Mechanical Engineering | Hasegawa Lab, The University of Tokyo

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Profile

- Commercial CFD and design optimization. Multiphysics for aerospace propulsion (fluid-structure coupled thermal, combustion coupling, supercritical fluids) on HPC clusters, plus topology and parametric optimization, across ANSYS Fluent, OpenFOAM and ToffeeX.
- Master's thesis drove a turbulent pipe-flow rig by Bayesian optimization with no operator in the loop: 75% drag reduction, 40% power saving rate. JSAE Graduate Research Encouragement Award.
- In Japan on a work visa. Japanese with clients, English for research, Chinese first language.

Education

The University of Tokyo

Oct 2023 – Sep 2025

Master, Mechanical Engineering, Hasegawa Lab

Tokyo, Japan

- Automatic Optimization of Pulsating Waveform for Drag Reduction in Experiment of Turbulent Pipe Flow
- Built an unattended Bayesian-optimization loop over the rig, reaching **75% maximum drag reduction** and a **40% maximum power saving rate**

Thesis received the JSAE Graduate Research Encouragement Award

University of Melbourne

Jan 2019 – Dec 2019

Master, Mechanical Engineering (Withdrawn)

Melbourne, Australia

- Investigation of energetic motions in rough-wall turbulent boundary layers

Withdrew on receiving a fully funded PhD offer from the same university; enrolment was deferred under COVID-19 travel and visa restrictions and the offer lapsed in late 2022 under university policy

Shanghai Jiao Tong University

Sep 2014 – Sep 2018

Bachelor, Energy and Power Engineering; Minor in Musicology

Shanghai, China

- Dilution effects of DME and ethanol on methane flames
- Dynamic characteristics of hydrogen-cooled generator stator foundations

Work Experience

SOLIZE Partners

Apr 2026 – Present

CAE Analysis Engineer

Kanagawa, Japan

- Fluid-structure coupled thermal, combustion-coupled and supercritical-fluid analysis for aerospace propulsion, run in parallel on an HPC cluster
- Delivered optimization PoCs: piping support placement (heat loss, thermal stress, seismic natural frequency) and coupled fluid-thermal-structural with topology optimization
- Cross-validation across ANSYS Fluent / OpenFOAM / ToffeeX; built the internal OpenFOAM computing environment (Slurm)

Independent Academic Tutor

2020 – 2023

AP / IB / A-Level Mathematics and Physics

Shanghai, China

- Taught students preparing for overseas university admission while reaching business-level Japanese and securing University of Tokyo admission

Skills

- CFD & Multiphysics**
ANSYS Fluent, OpenFOAM, ToffeeX
Conjugate heat transfer, fluid-structure interaction, combustion coupling, supercritical fluids, turbulence modelling
- Optimization**
Topology, parametric and hybrid optimization; Bayesian optimization with Gaussian-process surrogates; design of experiments
- Computing**
Python, C/C++, MATLAB
HPC clusters, Slurm, Git, GNU/Linux
- CAD & Workflow**
SolidWorks, CATIA, SpaceClaim
Meshing, boundary conditions, post-processing, validation, reporting

Languages: Mandarin Chinese (native) English (native level) Japanese JLPT N1 (business level)